

VibraHand: In-Hand Superpower Enabling Spying, Precognition, and Telekinesis

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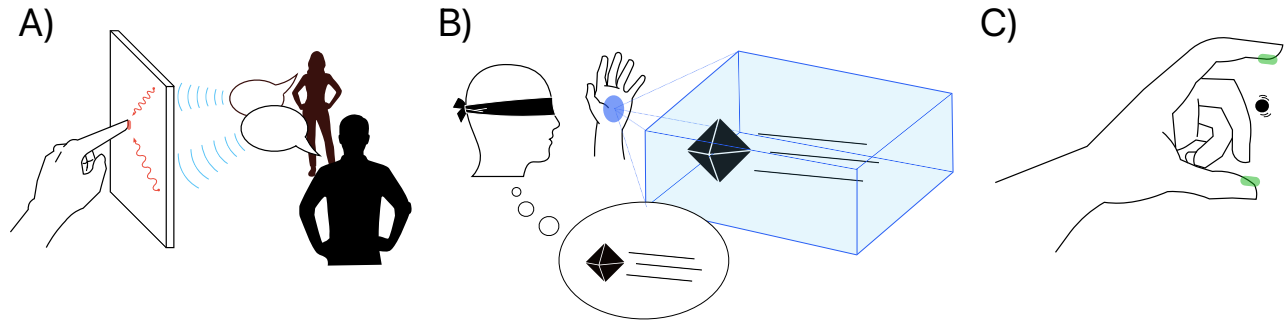


Figure 1: Three in-hand superpowers that VibraHand provides. Users can listen to sounds or events on the other side of a barrier through surface vibration sensing (A), react to nearby moving objects without seeing them using sonar and mmWave sensing (B), and control and manipulate small objects in the air with acoustic levitation (C).

ABSTRACT

This work presents a novel integration of multiple sensing and control technologies into a hand-worn device, enabling users to experience superpowers such as remote eavesdropping, telekinesis, and precognition. By leveraging techniques such as surface vibration sensing, ultrasound, and mmWave radar, the device facilitates expressive and intuitive in-hand interactions. Additionally, the use of acoustic levitation for contactless object manipulation extends the scope of wearable interactions. This innovative approach enhances the functionality of wearable devices for the future of seamless and powerful user experiences.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices; Interaction techniques.**

KEYWORDS

Surface acoustic wave, Active sonar, Millimeter wave, Acoustic levitation, Wearable interactions, Superpower

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1 INTRODUCTION

Wearable technologies have significantly advanced the capabilities of humans to sense and act, surpassing (or altering) our natural human abilities through augmentation by advanced technology. These developments allow people to experience future interactions, akin to experiencing *superpowers* [2, 13] such as flying in virtual reality [17] or extrasensory human perception that enables detecting invisible Wifi signals [13]. Such systems often incorporate external sensors in wearable form factors to facilitate on-body interactions based on regions such as head [1, 9], eye [17], arms [13] and hands [8, 14]. Among these, hand-based interactions are particularly compelling due to their prevalence in daily life; we very commonly interact with our hands, so equipping them with new abilities may have the most impact. Reflecting this perspective, much prior work has sought to enable more expressive hand-based interactions. Work has, for example, explored various around-body sensing techniques using signals in different frequency ranges such as surface vibrations (below 6KHz) [11, 19], ultrasound (sound in inaudible ranges, above 18KHz) [12, 16], and mmWave radar (above 24GHz) [20]. Beyond such sensing technologies, other work has explored the potential of new actuation technology that can enable us to control physical objects in mid-air, a seemingly magical and infeasible interaction. Such capabilities can be achieved by, for example, acoustic levitation using ultrasound, a technique that allows the manipulation of objects such as polystyrene beads [10, 14, 15], fabric [3, 15], or liquid [4]. GauntLev [14] showed the feasibility

of wearable designs in this space by proposing a worn device for contactless manipulation of dangerous or fragile materials.

In this work, we propose *VibraHand*, integrating multiple sensing and control channels across different frequency range signals into a single hand-worn form factor, enabling users to experience advanced interaction capabilities such as eavesdropping on speech via surface vibrations, precognition with radar and sonar sensing, and telekinesis using the mechanical energy of ultrasound. This novel combination enhances the expressiveness and functionality of wearable technology for future applications in human-computer interaction.

2 VIBRAHAND

We propose a hand-worn device that enables advanced interaction experiences by sensing and manipulating signals across diverse frequency ranges. Through a combination of surface vibration sensing, ultrasound, and mmWave radar, our system allows for experiences akin to superpowers, including spying, recognition, and telekinesis (see Figure 1). Below, we illustrate how each superpower is achieved with our system.

- **Spying** involves skills such as tailing, surveilling, and eavesdropping without being detected. We designed an advanced eavesdropping ability using surface vibration sensing to predict events [7] or listen to conversations [21] through walls. A single tiny voice-pickup unit is attached to the tip of the index finger. By simply placing the index finger against a wall, the user can listen to surface acoustic sounds, enabling them to hear conversations or predict events on the other side.
- **Precognition** is a knowledge of a future event or situation, especially through extrasensory means. These advanced sensing abilities can be designed using mmWave and sonar sensing. For example, human presence can be detected using a 24GHz mmWave sensor [5, 6], which can identify both moving and stationary states. Fine-grained nearby movements can be detected acoustically by emitting ultrasound and analyzing reflections using a single speaker and multiple microphones [16, 18]. These sensors are located on the palm to provide extrasensory spatial awareness.
- **Telekinesis** is an ability to move or deform inanimate objects in a contactless manner using hand gestures and mental focus. This capability can be achieved through acoustic levitation, which traps small objects or liquids using the mechanical energy of ultrasound. A pair of multiple ultrasonic transducers is located at the fingertips of each thumb and middle finger. By adjusting the distance between the transducers, the user can alter the distribution of ultrasound forces to trap and manipulate small objects [14].

3 IMPLEMENTATION

VibraHand fully utilizes the Gen-M Kit provided by the UIST Student Innovation Contest, including microcontrollers, an AI module, a mmWave sensor, and ultrasonic transducers. Additionally, we plan to incorporate extra sensors (e.g., voice pick-up units, microphone arrays, a speaker, and haptic actuators) and employ fabrication techniques (e.g., 3D printing). To enable in-hand superpowers, we

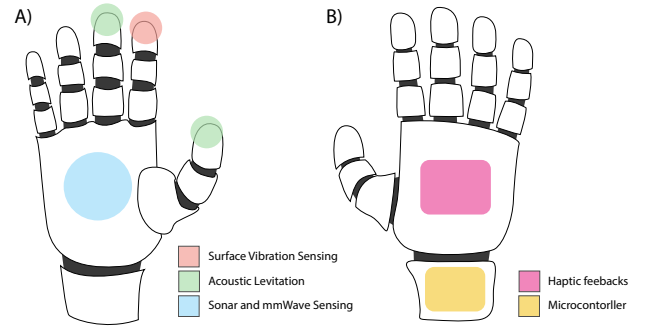


Figure 2: VibraHand design. A 3D-printed glove contains sensors and actuators enabling superpower on fingers and palm (A), and others (e.g., haptic actuator and microcontroller) on the back sides (e.g., back of the hand or wrist) (B).

designed a glove form factor, placing each sensor and actuator at the fingertips or palms as illustrated in Section 2. In addition, we placed haptic actuators at the back of the hand to assist spatial awareness during precognition, and others such as microcontrollers and batteries at the wrist (see Figure 2).

To have participants experience in-hand superpowers, we designed a series of tasks that fully utilize their in-hand sensing and control powers. First, blindfolded participants need to find a target box using precognition assisted by haptic feedback. Next, they can spy on the sound of clues whispering inside the target box. Using these clues, participants can locate and eliminate the final target through contactless manipulations with telekinesis. In addition to these gamified tasks, participants can also experience individual sensing and actuation techniques in more typical Human-Computer Interaction scenarios, such as gesture recognition using surface acoustic waves, hand movement tracking using sonar, and acoustic levitation.

4 CONCLUSION

This work introduces a hand-worn device that enables superpower-like experiences such as spying, telekinesis, and precognition. By integrating surface vibration sensing, ultrasound, and mmWave radar, our system offers intuitive and expressive in-hand interactions. Acoustic levitation for contactless object manipulation further expands the capabilities of wearable technology. This innovative approach advances human-computer interaction and opens new possibilities for future applications in various fields.

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